LinguaLore Project Report

Design for Educational Applications

CSCI 4080U

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Github Link: https://github.com/WarrenWarrens/EADFinalProject

Introduction

Literature, films, video games, and television shows that take places in worlds outside our own often create fictional languages as part of the setting. These help to give more depth to world building. Fans of these media often have an interest in learning these languages, as a way to engage with the stories on a deeper level. However, there is rarely a way for these fans to learn and practice these languages if they don't have like minded peers or know where to find the resources to help. Our app seeks to provide users with a way to learn these constructed languages, with activities that help them practice their reading, writing and speaking skills in various languages found in media.

Background / Related Work

Arguably the most popular & well known language learning app, "Duolingo", has started to break into the niche category of fictional languages with "Klingon" in 2017, & "High Valyrian" in 2018 [1]. However, other than an update to "High Valyrian" in 2022 to coincide with the release of the new "Game of Thrones" entry, "House of Dragons", there have been no major efforts or improvements into this niche category of language learning.

One question that comes up with discussing learning fictional languages is what purpose does it serve? Other than being fun, one benefit is that it helps you develop what is known as metalinguistic awareness, which is "the ability to consciously think about different properties of a language such as pronunciation and grammar"[2]. The creators of these fictional languages consciously design them, developing words and sentence structure entirely from scratch [3]. Developing this ability can help when learning other languages.

Learning a language requires interactive communication [4]. It’s not enough to simply understand the meaning of words you’re using, you also need to know how to use them and in what context. The usefulness of translation when teaching foreign languages has long been brought into question [5]. Therefore, a language learning app should have learning objects that focus on both translating words and practicing using them.

Feedback is another important strategy for language learning [6]. This involves identifying errors a learning is making and helping them to make changes in response. Learning objects should implement this by highlighting when a user gets something wrong in an activity.

One barrier to language learning comes from the fear of making mistakes [7]. Language learners fear they will be judged for making mistakes, treating them as something to be avoided rather than a vital part of the learning process [8]. Therefore, it's beneficial for learning objects to explain why they got something wrong and give users a chance to correct their mistakes, rather than simply punishing them for getting a question wrong.

Learning Outcomes

“LinguaLore” has 4 main learning outcomes. The first is that by the end of their selected language course, users will be able to understand conversations in the language without the use of in movie/tv show subtitles. We want users to be able to hear and listen to the language being spoken, and understand what is being said without having to look at the subtitles on screen. In the real world, there are no subtitles when

you are hearing someone else speak to or around you. We want the users to be able to understand as if they were totally immersed in that fictional environment.

The second learning outcome is that by the end of their selected language course, users will be able to formulate sentences and have a conversation with another language speaker to their desired fluency level. One of the critiques of “Duolingo" is that it is not effective at teaching proper grammar rules that one would be able to apply in a real conversation. The app is more focussed on teaching vocabulary with no context behind its words or sentences [9]. Research shows that focussing on grammar when teaching a language improves the overall effectiveness of teaching that language [10]. Our app aims to include grammar in our lessons so users feel confident in their abilities to form their own sentences and have meaningful conversations in their desired fantasy language.

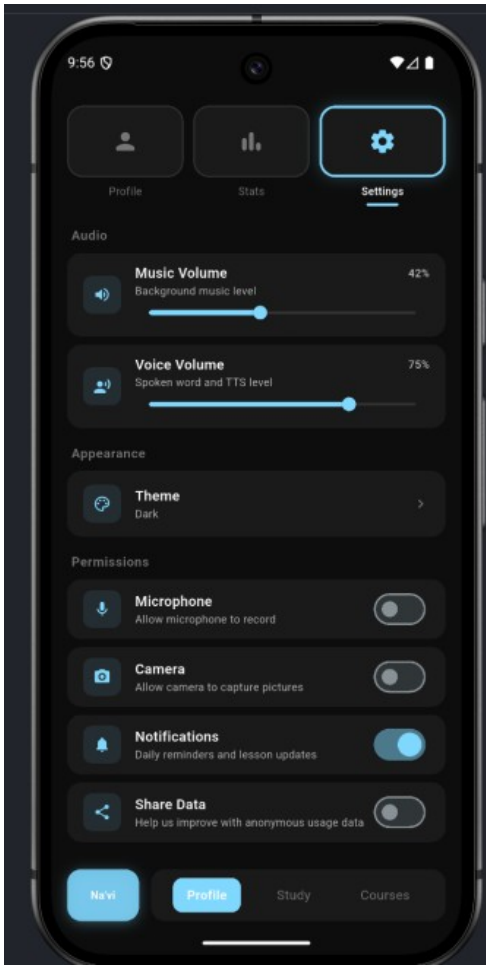
Our third learning outcome is that by the end of each lesson/module, users will be able to identify words and translate them to English. “LinguaLore” is designed so that at the end of each lesson, users will have been introduced to a new set of words or phrases. If they are not able to understand what those words or phrases mean in its English context, then our app has not done its job.

Our final learning goal is that by the end of each activity, users will be able to match their selected language words/phrases to its correct English meaning.

App Design

From the beginning, this application is made with the core belief that it needs to be as intuitive, non-invasive & fun. Starting from the beginning of launching the app for the first time as a new user, you are given three options, to sign into an existing account,

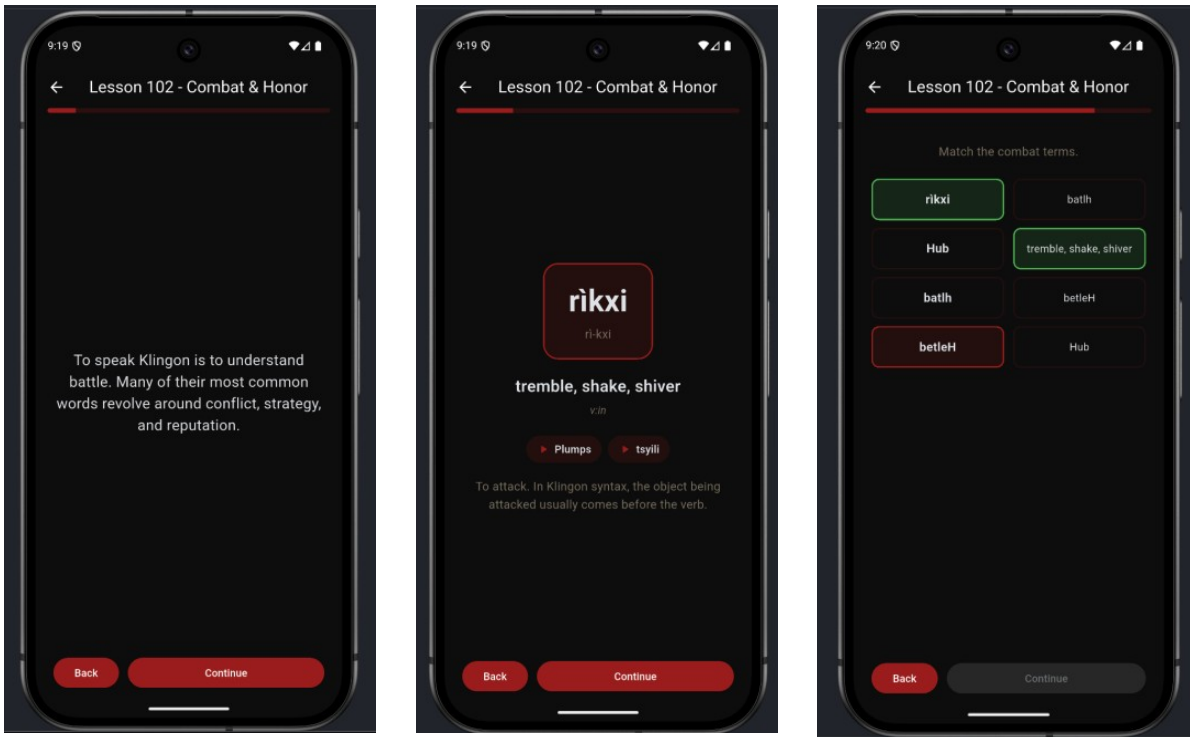
to create a new account or to continue as a guest. The first two options are the standard among any software, website or application, but giving the user the ability to still use the content of the application without needing an account, was an important feature we want to have, as it gives the user more freedom and allows them to focus more on what matters, that being actually interacting with the language learning activities. All settings they select at this point can be changed at any time within the user's profile.



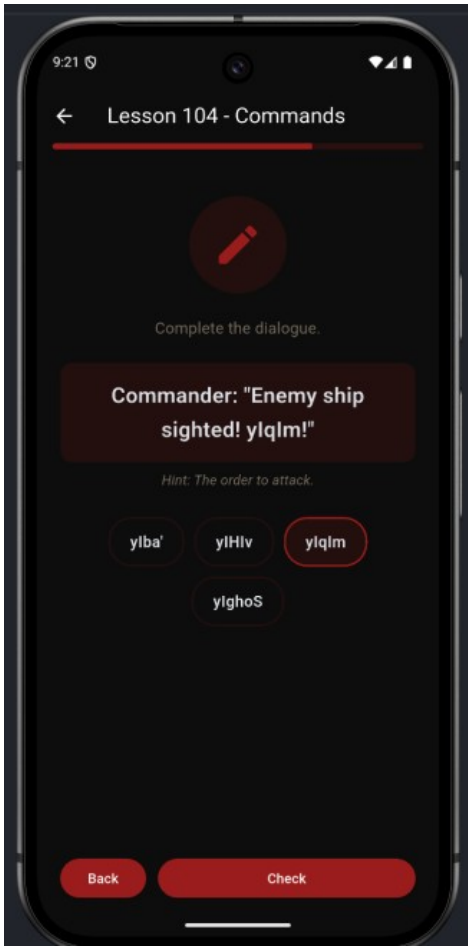
Profile Page

The main learning objects our app will feature are lessons, audio mimicry and conversation simulations. The lessons have the user go through slides, learning a specific content before attempting a short quiz at the end to test their knowledge. These

quizzes can take the form of matching games, multiple choice quizzes, or complete the sentence games. To build these lessons we used various online dictionaries organized by fans [11, 12, 13. 14. 15. 16].



Sample Lessons featuring teaching slides and matching game



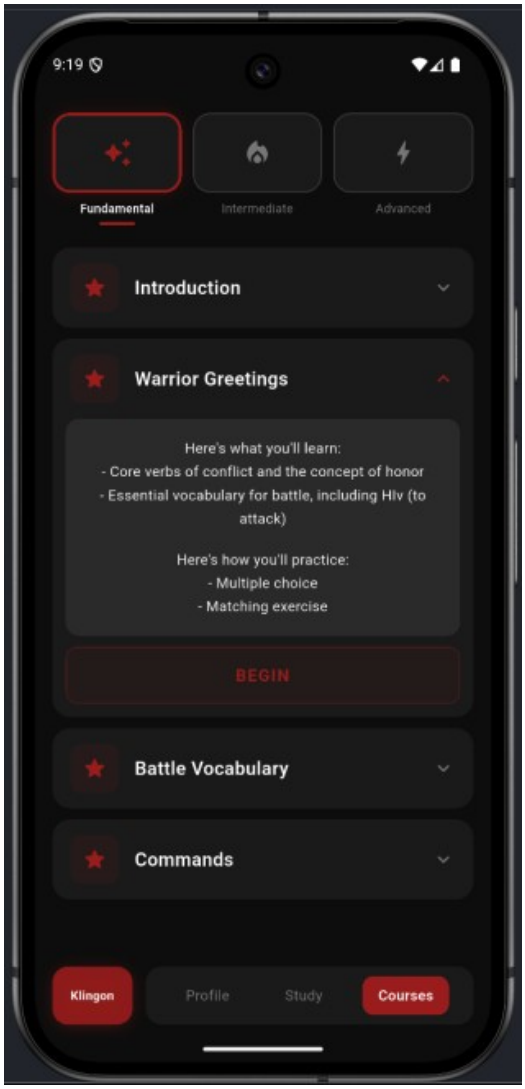
Complete the sentence game

In the audio mimicry activities, the user is presented with a word, featuring its phonetic pronunciation, trained via an online dictionary [17], and English translation. The user can then listen to the pronunciation and they try it for themselves. When the user doesn't get the pronunciation correct, the app highlights which portion they're having trouble with specifically. Once they've pronounced it correctly, they'll move on to the next one.

In the conversation simulations, the user can communicate via text with an LLM-powered tutor. Users type in either their desired fantasy language or english, and the tutor responses in their desired language, along with the english translation. Users

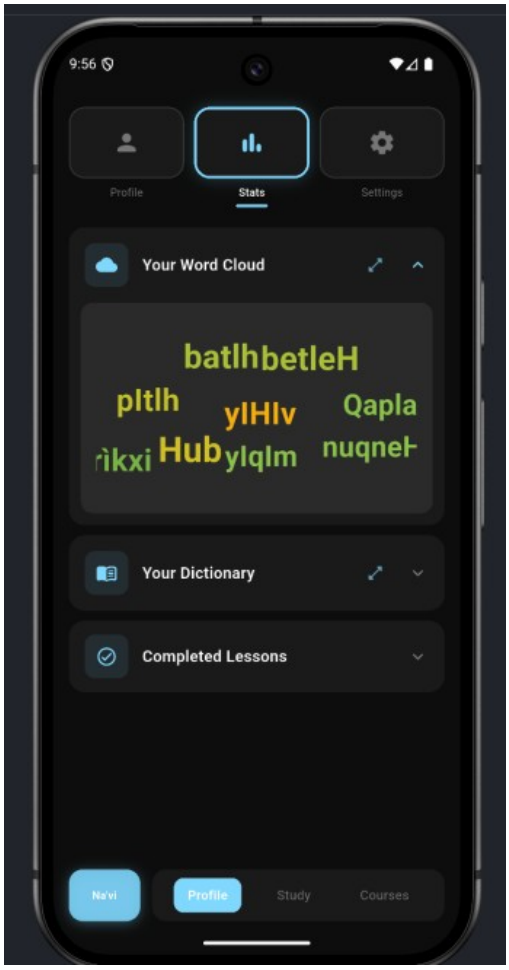
can hear the tutor's response via the text-to-sound feature. These learning objects are featured throughout the various lessons, giving users a variety of ways to learn the content.

Lessons are divided into three categories: Fundamental, Intermediate, and Advanced, with each lesson building on top of the previous ones. Before starting a lesson, a screen explaining what you'll learn and how you'll practice that lesson is displayed, ensuring users understand the expected outcome.

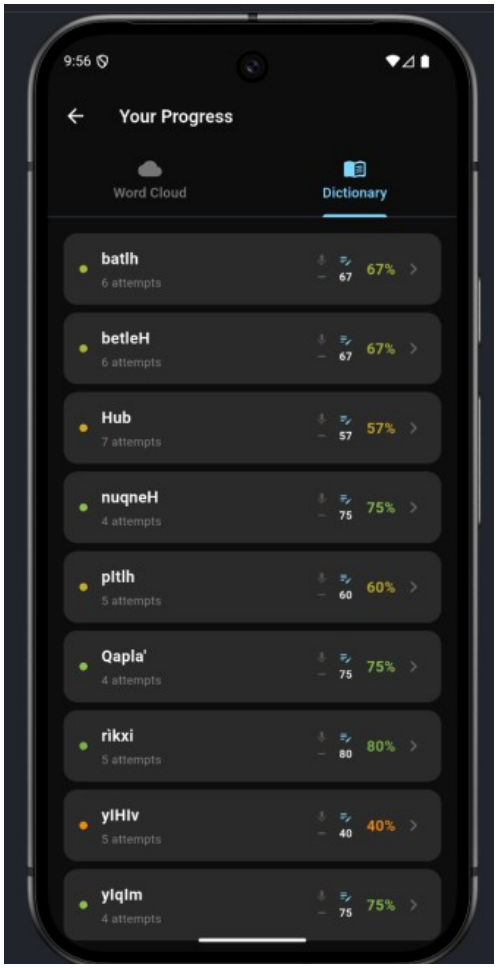


Lesson Description

Within the stats section of the app, users can view their progress. This is presented by a word cloud, which gives users a visual representation of their progress. The size of the word represents their word proficiency, with the larger words being the ones with the least amount of proficiency. This section also features a dictionary, where users can search for words. They can also click on the words to see their vocal, non-vocal, and overall proficiency.



Word Cloud



Dictionary

Evaluation

The purpose of the evaluation of our app would be to ensure LinguaLore is both enjoyable to use and educational. To this, we would conduct two sets of user studies. The first would mainly focus on LingaLore’s usability. We would have a group consisting of fans of Avatar and Star Trek use our app and then use a combination of surveys and interview them to learn about their experience. We would focus on how our app meets the usability heuristics criteria [18], and well as any pain points the users encountered while using LinguaLore.

Our second study would focus on LingauLore's educational value. First, we would

have participants complete a short quiz to gauge their proficiency in either Na’vi or Klingon prior to the app use. We would then have them use LinguaLore for a total of 4 30 minute sessions. At the end of the final session, the participants would do another quiz to gauge if they learned anything. We would also work with educators to ensure our app was as effective as possible.

Conclusion

LinguaLore is a language learning app that focuses on teaching its users the fantasy language found in shows and movies. By combining educational and entertaining elements, LinguaLore seeks to bridge the gap between the real world and the fantastical. In the future LinguaLore could be expanded to cover more fantasy languages. Na'vi and Klingon were chosen because of the breadth of online resources available regarding these languages. However, that doesn't mean there isn't a desire for people to learn these lesser documented languages. Further fine-tuning of the LLM powered features could also be implemented, as well as a daily quiz feature to encourage users to return.

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